EXPERIMENT-7b

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Develop Low-Fidelity Paper Prototypes for a Banking App and Convert Them into Digital Wireframes Using Inkscape

AIM

The aim of this experiment is to construct low-fidelity paper prototypes for a banking application and digitize them into wireframes using Inkscape, a free and open-source vector graphics editor. The experiment covers the complete UI wireframing workflow from hand-drawn sketches to structured digital wireframes with proper shapes, labels, and navigation indicators.

Tools Used

- Inkscape – for creating digital vector wireframes
- Paper and Pencil – for initial low-fidelity sketches
- Tool Link: https://inkscape.org/

Procedure

Step 1: Create Low-Fidelity Paper Prototypes

1. Identify Core Features:

The banking application is designed to allow users to manage their accounts, perform transactions, and access banking services. The following core features and screens were identified:

- Login Screen – Mobile number / username and MPIN entry with biometric options.
- Dashboard – Account balance tabs (Savings/Current/FD), quick actions, and recent transactions.
- Account Details – Account summary card, account statistics, and mini statement.
- Fund Transfer – Transfer type tabs (NEFT/RTGS/IMPS/UPI), beneficiary details, amount, and IFSC.
- Bill Payments – Service category grid and saved billers with due amounts.
- Settings & Security – Profile summary, security settings, preferences, and support menu.

2. Sketch Basic Layouts:

Plain paper and a pencil were used to sketch the basic layout of each screen. The focus was on placement of primary elements — buttons, input fields, navigation menus, and cards — without worrying about colours or visual styling. Rough boxes and labels represented all UI components.

- Simple rectangles were drawn for input fields, cards, and buttons.
- Navigation bars were sketched at the top and bottom of each screen.
- Labels and placeholder text were written inside elements to describe their function.
- Arrows were drawn between screens to indicate navigation flows.

3. Iterate and Refine:

The paper sketches were reviewed with peers to gather feedback before converting to digital wireframes. The following improvements were made based on feedback:

- Dashboard: Added account type tabs (Savings / Current / FD/RD) at the top for multi-account users.
- Fund Transfer: IMPS charges info bar added below the form for transparency.
- Bill Payments: Saved billers section added at the bottom to enable faster repeat payments.
- Settings: Grouped menu items under clear headings (Security, Preferences, Support) for easier navigation.

Step 2: Convert Paper Prototypes to Digital Wireframes Using Inkscape

1. Install Inkscape:

Inkscape was downloaded from the official website (https://inkscape.org/) and installed. A blank document was created to begin wireframing.

2. Create a New Document:

A new document was created in Inkscape using File > New. A mobile canvas size of 360 × 640 pixels was set to match standard smartphone screen dimensions.

3. Set Up the Document:

- Document dimensions were set via File > Document Properties (360 × 640 px).
- A grid was enabled via View > Page Grid to assist with element alignment.
- Snapping was turned on to ensure elements aligned to grid lines consistently.

4. Draw Basic Shapes:

Inkscape's rectangle and ellipse tools were used to draw all UI elements. The wireframe style used thin black outlines with light grey fills to represent the interface in a neutral, uncoloured format:

- Rectangle tool (R): Used for input fields, buttons, cards, navigation bars, and containers.
- Ellipse tool (E): Used for avatar placeholders, icon circles, and toggle buttons.
- Line tool: Used for dividers, underlines, and navigation flow arrows.
- Bezier/Pen tool: Used for custom icons and arrow connectors between screens.

5. Add Text:

The text tool (T) was used to add labels, headings, placeholder text, and annotations to every element. Font size 10–14px was used for labels; bold 16px for screen titles and section headings.

6. Organise and Align:

- Elements were grouped by screen section using Object > Group (Ctrl+G).
- The Align and Distribute panel (Shift+Ctrl+A) was used to align elements horizontally and vertically.
- Equal spacing was applied between repeated elements such as form fields and list items.

7. Group Elements:

Related UI components on each screen were grouped together to keep the canvas organised. For example, the navigation bar, the balance card, and the transaction list were each grouped separately. This made it easy to move and edit each section independently.

8. Create Multiple Screens:

Six screens were created for the SecureBank Banking Application by duplicating the base layout using Edit > Duplicate (Ctrl+D) and modifying the content for each screen:

- Screen 1: Login Screen – Entry point with MPIN login and biometric options.
- Screen 2: Dashboard Screen – Account balance, quick actions, and recent transactions.
- Screen 3: Account Details Screen – Full account info card and mini statement.
- Screen 4: Fund Transfer Screen – NEFT/RTGS/IMPS/UPI transfer with beneficiary form.
- Screen 5: Bill Payments Screen – Category grid and saved billers with quick Pay Now.
- Screen 6: Settings & Security Screen – Security settings, preferences, and support menu.

9. Link Screens (Optional):

Navigation flow arrows were added between screens using Inkscape's Bezier tool. Arrows were drawn with labels to indicate user actions that trigger navigation:

- Login button → Dashboard
- Open an Account link → Registration Screen (future scope)
- Transfer icon on Dashboard → Fund Transfer Screen
- Pay icon on Dashboard → Bill Payments Screen
- Account balance card on Dashboard → Account Details Screen
- More tab in navigation bar → Settings & Security Screen
- Back button on all sub-screens → Dashboard

10. Export Wireframes:

Each screen was exported as a PNG image using File > Export PNG Image (Shift+Ctrl+E). Each page was selected individually, the export area was set to "Page", and a 96 dpi resolution was applied. The exported files were saved for documentation.

Digital Wireframes – Output Screens

The following screenshots show the six digital wireframes created in Inkscape for the banking application:

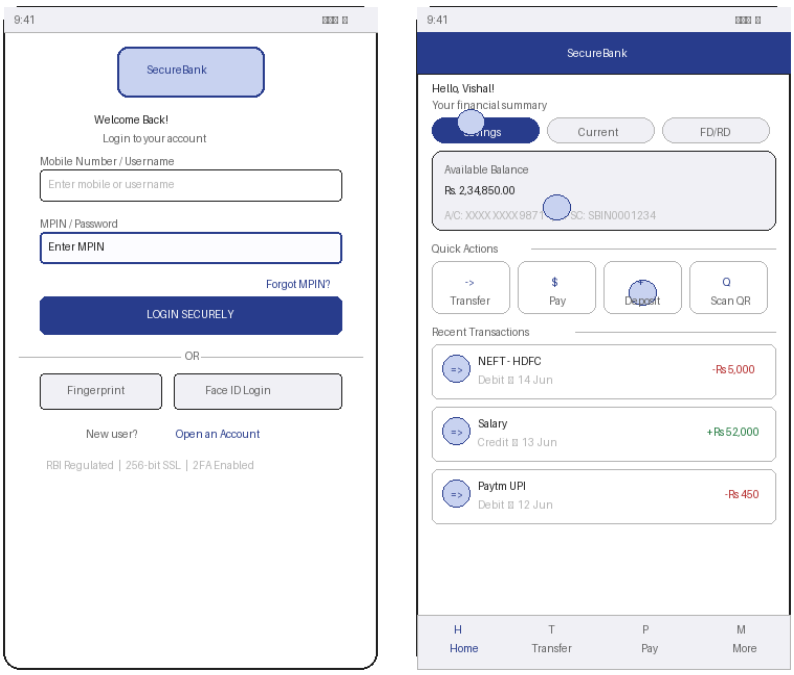

Fig 1: Login Screen (left) | Dashboard Screen (right)

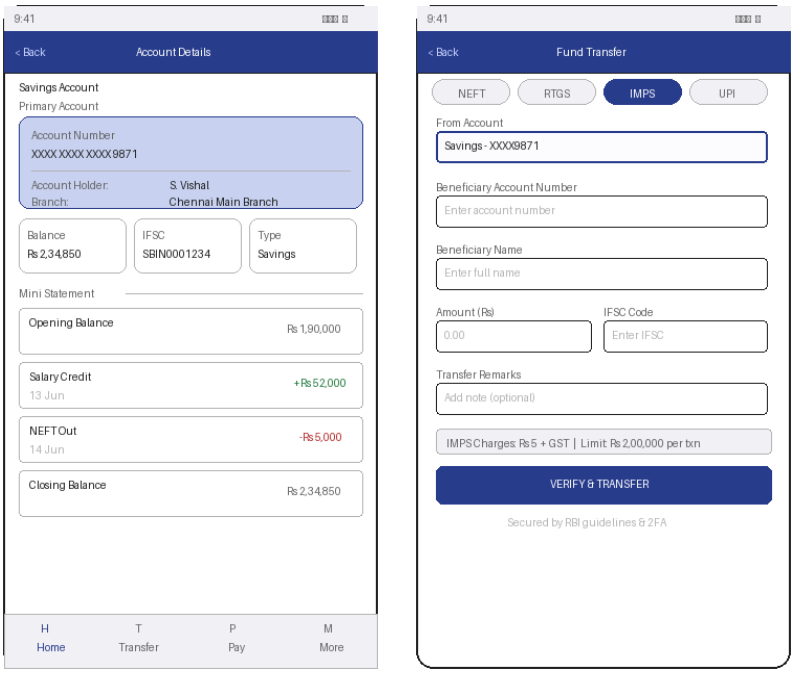

Fig 2: Account Details Screen (left) | Fund Transfer Screen (right)

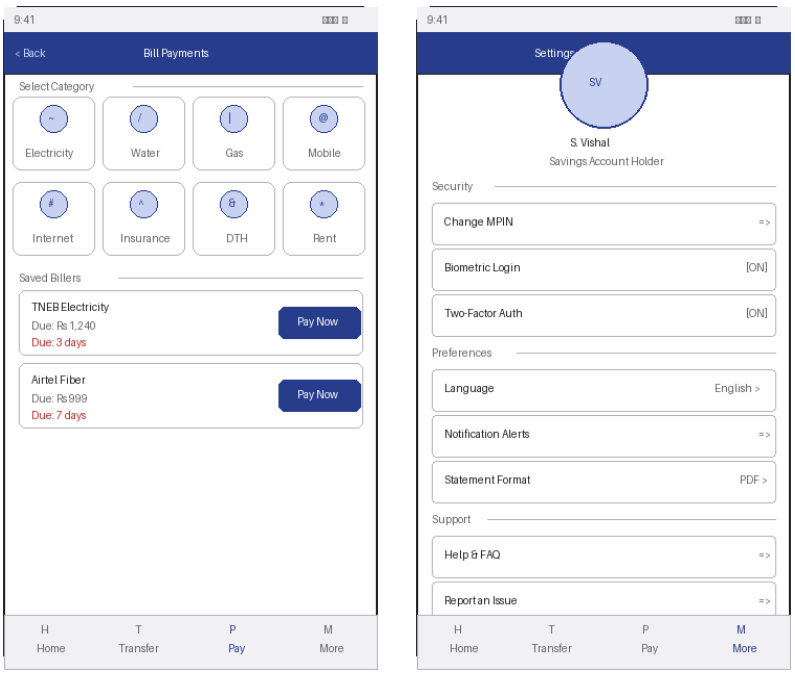

Fig 3: Bill Payments Screen (left) | Settings & Security Screen (right)

Screen Descriptions

Screen 1: Login Screen

The Login screen serves as the secure entry point for the SecureBank application. It displays the bank logo at the top, followed by input fields for Mobile Number/Username and MPIN/Password. A "Forgot MPIN?" link is provided for account recovery. The primary "Login Securely" button is prominently placed. Below a divider, two alternative login options — Fingerprint and Face ID — are shown as icon buttons. A link at the bottom navigates new users to the account opening page. A trust badge line shows RBI regulation and SSL encryption indicators.

Screen 2: Dashboard Screen

The Dashboard is the primary home screen after authentication. A personalised greeting is displayed at the top. Account type tabs (Savings, Current, FD/RD) allow switching between multiple accounts. A balance card below shows the available balance, masked account number, and IFSC code. A Quick Actions row provides shortcuts to Transfer, Pay, Deposit, and Scan QR. The Recent Transactions section lists the latest transactions with icons, names, types, dates, and colour-coded amounts (green for credit, red for debit). A persistent bottom navigation bar provides access to Home, Transfer, Pay, and More.

Screen 3: Account Details Screen

The Account Details screen provides a comprehensive view of the selected account. A highlighted account card shows the full masked account number, account holder name, and branch name. Below it, three statistics tiles display the current balance, IFSC code, and account type. A Mini Statement section lists opening balance, recent credits and debits, and closing balance in a table-style layout with colour-coded amounts. This screen helps users verify account information and review a quick financial summary without navigating to the full transaction history.

Screen 4: Fund Transfer Screen

The Fund Transfer screen allows users to send money using four transfer modes: NEFT, RTGS, IMPS, and UPI, selectable via tab buttons at the top. The form includes fields for From Account (pre-filled), Beneficiary Account Number, Beneficiary Name, Amount, IFSC Code, and optional Remarks. Amount and IFSC fields are placed side-by-side to reduce vertical scrolling. A charges information bar below the form shows applicable IMPS charges and transaction limit. The "Verify & Transfer" button initiates the transfer flow, and a security note confirms encryption.

Screen 5: Bill Payments Screen

The Bill Payments screen is divided into two sections. The top section displays an 8-icon category grid covering Electricity, Water, Gas, Mobile, Internet, Insurance, DTH, and Rent, allowing users to select a biller category. The bottom section lists Saved Billers — previously paid billers from the user's history — showing the biller name, due amount, and due date in red for urgency. Each saved biller has a "Pay Now" button for direct one-tap payment, reducing the steps needed for repeat bill payments.

Screen 6: Settings & Security Screen

The Settings & Security screen is organised under the profile summary at the top, which shows the user's initials, name, and account type. Three grouped sections follow: Security (Change MPIN, Biometric Login toggle, Two-Factor Auth status), Preferences (Language selection, Notification Alerts, Statement Format), and Support (Help & FAQ, Report an Issue, Contact Bank). Each menu item displays a label on the left and a value or arrow indicator on the right. A red "Logout" button at the bottom clears the session and returns the user to the Login screen.

Result

The experiment was completed successfully. Low-fidelity paper prototypes were created for six screens of the SecureBank Banking Application: Login Screen, Dashboard Screen, Account Details Screen, Fund Transfer Screen, Bill Payments Screen, Settings & Security Screen. These paper sketches were then converted into structured digital wireframes using Inkscape. All screens were drawn using Inkscape's shape and text tools, grouped for organisation, annotated with labels, linked with navigation arrows, and exported as PNG images for documentation.

Conclusion

This experiment demonstrated the complete wireframing workflow using Inkscape as a vector-based design tool. Starting from low-fidelity paper sketches allowed rapid exploration of layout ideas without any technical overhead. Inkscape proved to be a capable tool for converting those sketches into well-structured digital wireframes with precise alignment, grouping, and annotation. The SecureBank Banking Application wireframes effectively communicated the interface structure and navigation flow, validating the design before any development effort begins. This approach reduces usability issues early in the design process, saving time and cost in later development stages.